

STRATEGIC ANALYSIS

**ANALYZING THE IMPACT OF DIGITAL BANKING EXPERIENCE
ON CUSTOMER LIFETIME VALUE AND CUSTOMER RETENTION
RISK AT BANK CIMB NIAGA**

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Executive Summary

Digital banking has become a core driver of customer engagement, retention, and long-term value creation at Bank CIMB Niaga, with Octo Mobile serving as the primary interface between the bank and its customers. As customer interactions increasingly shift to mobile platforms, the quality and reliability of the digital banking experience now play a decisive role in shaping customer trust, usage behaviour, and overall lifetime value. Consequently, persistent digital experience issues represent not only operational challenges but also material business risks.

This study analyses 45,524 user reviews of the Octo Mobile application from the Google Play Store, focusing on low-rated feedback (1–3 stars) as early indicators of customer dissatisfaction and retention risk. Sentiment analysis reveals that 57.4 percent of reviews are negative, highlighting systemic weaknesses in the digital banking experience rather than isolated incidents. Login and access issues emerge as the most dominant pain point, followed by feature malfunctions, UI/UX complexity, performance instability, and transaction errors. These issues consistently occur at critical moments in the customer journey, particularly during authentication and transaction execution.

From a business perspective, these digital experience gaps translate into significant Customer Lifetime Value exposure. Based on conservative assumptions, the analysis estimates a total CLV at risk of approximately Rp 33.5 billion, driven primarily by login and access failures, transaction and transfer errors, and unstable core features. These issues directly increase customer churn probability, reduce transaction frequency, and limit cross-selling opportunities, particularly among digitally active and high-value customers.

Strategically, the findings position digital banking experience as a core revenue protection and competitive differentiation lever for Bank CIMB Niaga. Persistent digital friction erodes customer loyalty, weakens brand trust, and lowers switching barriers in an increasingly competitive digital banking landscape. Without targeted intervention, these challenges pose a sustained threat to long-term profitability and CLV sustainability.

To mitigate these risks, the report recommends a value-driven remediation approach that prioritizes initiatives based on CLV impact rather than operational convenience. Immediate stabilization of login, authentication, and transaction reliability is critical, supported by proactive retention strategies for high-value customers experiencing repeated disruptions. Success should be measured through CLV-linked and experience-driven metrics, including churn reduction, transaction success rates, sentiment improvement, and long-term CLV stabilization.

This analysis demonstrates that improving the digital banking experience is not merely an IT enhancement initiative but a strategic imperative to protect customer lifetime value, strengthen competitive positioning, and sustain long-term growth for Bank CIMB Niaga.

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Introduction

Digital banking has become a central pillar in the growth strategy of Bank CIMB Niaga, reshaping how customers interact with banking products and services. As customer behaviour increasingly shifts from physical branches to mobile platforms, digital channels now serve not only as transaction tools but also as primary drivers of customer engagement, retention, and cross-selling opportunities. In this context, Octo Mobile plays a critical role as the main interface between the bank and its customers, directly influencing daily usage patterns and long-term customer relationships. The quality and reliability of the digital banking experience therefore, hold strategic importance, as any friction within the mobile platform has the potential to affect customer trust, usage intensity, and ultimately the bank's ability to sustain long-term growth.

As digital banking becomes the primary channel through which customers engage with Bank CIMB Niaga, the success of this channel can no longer be evaluated solely through usage metrics or adoption rates. Instead, its strategic value must be assessed based on how effectively it supports long-term customer relationships and sustained profitability. In this context, Customer Lifetime Value represents a critical measure of long-term business performance, as most of the banking revenue is generated through continuous engagement rather than one-time transactions. High-value customers contribute not only through interest income but also via recurring fee-based products such as transfers, cards, and investment services. Retaining these customers and expanding their product adoption over time is therefore more valuable than continuously acquiring new customers. Any deterioration in digital experience has the potential to weaken engagement intensity, limit cross-selling opportunities, and ultimately increase the risk of losing customers with significant lifetime value.

As Customer Lifetime Value is highly dependent on sustained engagement over time, any disruption in the digital banking experience can translate into a measurable increase in customer retention risk. In mobile banking, issues such as application instability, failed transactions, or authentication problems do not merely create temporary dissatisfaction but can trigger behavioral changes that reduce usage frequency or prompt customers to consider alternative banking providers. Given the relatively low switching cost in digital financial services, negative digital experiences can accelerate churn, particularly among customers who rely heavily on mobile banking for daily transactions. User reviews and feedback therefore serve as an early indicator of retention risk, capturing signals of frustration and declining trust that may not yet be visible in traditional performance metrics but pose a direct threat to the sustainability of customer lifetime value.

Building on the relationship between digital banking experience, Customer Lifetime Value, and customer retention risk, the following strategic issues define the core business problem addressed in this report:

1. Digital banking has become the primary customer touchpoint at Bank CIMB Niaga, making the mobile experience a critical driver of customer retention and long-term value.
2. Friction within Octo Mobile increases customer retention risk, particularly among high-value customers with significant Customer Lifetime Value.

3. Customer reviews reveal gaps between customer expectations and the actual digital banking experience, signaling potential business risks.
4. These gaps may lead to reduced engagement, lower cross-selling effectiveness, and erosion of long-term revenue.
5. A data-driven approach is required to quantify how digital experience issues translate into Customer Lifetime Value exposure and retention risk.

Data and Analytical Approach

A. Data Sources and Scope

- **Data source:** User reviews of the Octo Mobile application collected from Google Play Store.
https://drive.google.com/file/d/16h_Pf1y-7SMw4_sL31zuu750BAMmzgmf/view?usp=sharing
- **Data type:** Unstructured text reviews, user ratings, review timestamps, and application version information.
- **Period of analysis:** Reviews collected over a defined multi-period timeframe to capture sentiment and experience trends.
- **Unit of analysis:** Individual customer reviews representing digital banking experience.
- **Scope of analysis:** Digital banking experience related to core functionalities, transaction reliability, security, and usability.
- **Data role:** Review data is used as a proxy for customer feedback and an early indicator of customer retention risk.

B. Text Analytics and Sentiment Classification

The text analytics and sentiment classification were conducted on 45,524 customer reviews of the CIMB Niaga mobile banking application, limited to ratings between one and three stars. This subset was intentionally selected to capture user experiences characterized by dissatisfaction, friction, or unmet expectations, as these segments represent the highest risk contributors to customer churn and declining Customer Lifetime Value (CLV). By focusing on this rating range, the analysis emphasizes early warning signals of retention risk within the digital banking ecosystem.

Sentiment classification categorized customer feedback into three classes: negative, neutral, and positive. The results indicate a strong dominance of negative sentiment, accounting for 57.36 percent of total reviews, followed by neutral sentiment at 33.85 percent, while positive sentiment represents only 8.80 percent. This imbalance highlights a structural gap between customer expectations and actual digital service delivery, suggesting that a large proportion of users experience persistent pain points rather than isolated issues.

Further analysis of sentiment distribution across rating levels reveals a clear gradient in customer perception. One-star ratings are heavily concentrated in negative sentiment, reflecting acute service failures and usability barriers. In contrast, three-star ratings exhibit a higher proportion of neutral and mildly positive sentiment, indicating that while some users are able to complete basic transactions, the overall experience lacks the reliability and convenience required to foster long-term loyalty.

To identify the primary drivers behind negative sentiment, frequent word and key phrase analysis was performed on negatively classified reviews. The analysis reveals three dominant issue clusters: access and login failures, non-functioning or unstable features, and UI/UX combined with performance-related problems. These issues are closely linked to critical moments in the customer journey, particularly during authentication, transaction execution, and post-update usage, which are essential touchpoints for sustaining engagement in digital banking platforms.

Table 1. Major Issue Categories Identified from Negative Reviews

Issue Category	Number of Mentions	% of Total Negative Reviews
Login & Access Issues	10,268	39.32%
Feature Not Working	8,046	30.81%
UI/UX Issues	7,694	29.47%
Performance Issues	5,735	21.96%
Transaction Errors	5,715	21.89%
Update Problems	4,836	18.52%
OTP & Verification	2,843	10.89%
Customer Service	1,193	4.57%

From a business perspective, these findings demonstrate that deficiencies in digital banking experience directly undermine customer confidence, reduce transaction continuity, and increase the likelihood of customer disengagement. Persistent exposure to such issues can shorten customer tenure, lower transaction frequency, and suppress monetary value per user, ultimately eroding CLV. Consequently, the outcomes of this text analytics and sentiment classification serve as a foundational input for subsequent CLV modeling and customer retention risk assessment.

Table 2. Issue Prioritization Based on Frequency and Severity

Priority Rank	Issue Category	Mentions	Severity Level
1	Login & Access Issues	10,268	Critical
2	Feature Not Working	8,046	High
3	UI/UX Issues	7,694	High
4	Performance Issues	5,735	High
5	Transaction Errors	5,715	High
6	Update Problems	4,836	Medium
7	OTP & Verification	2,843	Medium
8	Customer Service	1,193	Medium

Key Findings and Business Insights

Key Findings

- Customer sentiment from low-rated reviews (1–3 stars) is predominantly negative, with 57.4 percent of total reviews expressing dissatisfaction, indicating systemic weaknesses in the digital banking experience.

- Login and access issues represent the most dominant pain point, accounting for over 43 percent of negative reviews, highlighting critical failures at the initial stage of customer interaction with the app.
- Feature malfunction and transaction-related errors appear consistently across reviews, suggesting instability in core banking functionalities that directly affect daily financial activities.
- UI/UX complexity and application performance issues remain significant contributors to negative sentiment, reflecting a mismatch between user expectations and actual usability.
- Although positive sentiment exists, it is limited to 8.8 percent of total reviews and is largely associated with requests for fixes rather than sustained satisfaction, signaling weak advocacy and loyalty signals.
- The concentration of repeated keywords related to access, errors, and updates indicates recurring and unresolved issues rather than isolated operational incidents.

Business Impact on Customer Lifetime Value (CLV)

The analysis reveals that persistent digital banking issues pose a material financial risk to Bank CIMB Niaga through their direct impact on customer churn and Customer Lifetime Value. Based on conservative assumptions of an average CLV of Rp 5,000,000 per customer and elevated churn rates linked to negative digital experiences, multiple issue themes expose the bank to substantial potential revenue loss.

Login and access issues alone affect more than 11,000 customers and are associated with an estimated CLV exposure exceeding Rp 8.4 billion, making them the single largest contributor to value at risk. Transaction and transfer errors, while affecting a smaller user base, carry the highest churn rate at 25 percent, resulting in a CLV risk of over Rp 7.1 billion. Feature malfunctions and application performance instability further compound this exposure by limiting transaction frequency and reducing customer engagement with value-generating services.

Overall, the cumulative CLV at risk from identified digital experience issues is summarized below.

Table 3. CLV at Risk

Issue Theme	Affected Customers	Estimated Churn	CLV at Risk (IDR)
Login and Access Issues	11,236	15.0%	Rp8,427,000,000
Features Not Working	10,031	12.5%	Rp6,269,375,000
Transaction and Transfer Errors	5,721	25.0%	Rp7,151,250,000
App Performance and Stability	7,594	10.0%	Rp3,797,000,000
UI/UX Complexity	7,697	9.0%	Rp3,463,650,000
App Update Problems	4,836	7.5%	Rp1,813,500,000
OTP and SMS Verification	2,697	12.5%	Rp1,685,625,000
Customer Service Issues	1,203	15.0%	Rp902,250,000

Total CLV at Risk	Rp33,509,650,000
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From a strategic perspective, these findings position digital banking experience not merely as an operational or IT concern, but as a core revenue protection lever. Without targeted intervention on high-impact issues, particularly login reliability and transaction stability, Bank CIMB Niaga faces sustained erosion of long-term customer value and competitive positioning in the digital banking market.

Strategic Implications for Bank CIMB NIAGA

Persistent challenges in the digital banking experience carry direct and material implications for Bank CIMB Niaga's revenue and profitability. Repeated login failures, transaction disruptions, and unstable application performance reduce customers' ability and willingness to transact frequently, weakening fee-based income and limiting monetization of core banking services. Over time, these frictions suppress cross-selling potential for higher-margin products, positioning digital experience issues not as isolated operational problems, but as structural constraints on sustainable revenue growth.

Beyond immediate financial impact, prolonged digital friction gradually erodes customer loyalty and brand trust. In the current banking environment, customers increasingly equate digital reliability with institutional credibility. When fundamental digital interactions fail, confidence in the bank's overall service quality diminishes, lowering emotional attachment and increasing sensitivity to future service disruptions. As trust weakens, customers become less tolerant, more vocal in complaints, and less inclined to maintain long-term relationships with the brand.

This erosion of trust has direct competitive consequences. As switching barriers decline, dissatisfied customers are increasingly inclined to migrate toward alternative digital banking platforms perceived as more stable and reliable. Publicly visible negative app reviews further amplify this risk by shaping market perception, positioning digital experience not as a hygiene factor, but as a decisive competitive differentiator.

Collectively, these dynamics place long-term Customer Lifetime Value sustainability at risk, as ongoing digital experience issues disrupt the value creation cycle across acquisition, engagement, and retention. To safeguard CLV over time, the following strategic implications emerge:

- Sustained digital reliability is critical to preserving customer tenure and preventing premature churn.
- Consistent, frictionless experiences enable deeper product adoption and higher revenue per customer over time.
- Trust built through stable digital interactions strengthens long-term loyalty and reduces price and feature sensitivity.
- Improving digital experience quality protects existing CLV while creating a stronger foundation for future customer value growth.

Strategic Recommendations and Roadmap

To effectively mitigate CLV erosion and restore long-term value creation, Bank CIMB Niaga must adopt a value-driven approach that targets the primary drivers of customer churn while rebuilding digital trust. Rather than treating digital issues as isolated technical incidents, improvement efforts should be anchored in their impact on customer lifetime value, ensuring that resources are allocated to initiatives that meaningfully protect and expand long-term customer relationships.

Building on this value perspective, initiative prioritization must be guided by financial exposure rather than operational convenience. Login & Access Issues and Transaction & Transfer Errors represent the largest sources of CLV at risk, making the stabilization of these critical customer journeys an immediate strategic imperative. Strengthening reliability and transaction integrity at the core of the digital experience should precede cosmetic enhancements, as persistent failure in these areas directly accelerates churn, weakens trust, and constrains sustainable value creation.

Key priority initiatives include:

- End-to-end stabilization of login, authentication, and session management flows to eliminate recurring access failures.
- Hardening of transaction and transfer processes through real-time validation, rollback mechanisms, and clearer failure messaging.
- Reduction of OTP and SMS dependency risks by improving delivery reliability and introducing alternative secure authentication channels.
- Simplification of UI and navigation for high-frequency transactions to minimize cognitive and operational friction.

While systemic fixes are underway, targeted retention mechanisms are required to protect high-CLV customers who are most sensitive to repeated service disruptions. Retention should shift from reactive complaint handling to proactive experience protection.

Table 4. Recommended Retention Actions for High-Value Customers

No	Retention Action	Strategic Objective
1	Identify high-value customers experiencing repeated digital failures using behavioral and complaint signals	Early detection of churn risk among high-CLV customers
2	Proactive outreach and issue resolution via dedicated digital support channels	Restore trust and reduce frustration before churn occurs
3	Temporary fee waivers, transaction cost reimbursements, or loyalty incentives	Compensate service disruption and reinforce perceived value
4	Priority access to beta fixes or stabilized app versions	Rebuild confidence and increase tolerance during recovery period

The effectiveness of the strategy should be evaluated through CLV-linked and experience-driven metrics rather than technical outputs alone.

Table 5. Key Success Metrics for CLV Protection Strategy

Metric Category	Key Performance Indicator
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Customer Retention	Reduction in churn rate among digitally active and high-CLV customers
Customer Sentiment	Decline in negative and neutral app store reviews related to access and transaction issues
Digital Reliability	Increase in successful login and transaction completion rates
Customer Engagement	Improvement in monthly active users and transaction frequency per customer
Long-Term Value	Stabilization and growth of average Customer Lifetime Value over time